

Chapter 4 --- Database, Schema, Table, Row and Column

Relational databases organize data in a clear structure. Before creating a database, you need to understand the different levels used to store and arrange data.

The structure can be viewed as: MySQL Server → Database/Schema → Table → Rows and Columns → Values

4.1 Database

A database is an organized collection of related data. It acts as a main container that holds:

- Tables
- Relationships
- Views
- Indexes
- Other database objects

For example, a college may create a database called CollegeManagement, which can contain information about:

- Students
- Courses
- Trainers
- Attendance
- Marks
- Placements

All this information belongs to the same college application, so it can be stored inside one database.

4.2 Schema

A schema represents the logical organization or design of a database. It describes:

- What tables exist
- What columns each table contains
- What type of data each column stores
- How tables are connected
- What rules are applied to the data

A schema can be understood as the blueprint of a database.

For example, a college database schema may define a Students table, a Courses table, an Enrollments table, and the relationship between students and courses.

Schema in MySQL

In MySQL, the terms database and schema are generally used interchangeably. Therefore, when MySQL Workbench displays a section called Schemas, it is showing the databases available on the MySQL server.

4.3 Table

A table stores data about one particular type of entity. An entity is something about which data is stored. Examples include: Student, Employee, Product, Customer, Course. A table organizes data using rows and columns.

Consider a Students table:

StudentID	StudentName	Branch	City
-----	-----	-----	-----
101	Aarav	CSE	Bengaluru
102	Bhavna	ECE	Pune
103	Charan	CSE	Hyderabad

The table stores information about multiple students. A database can contain many tables, and each table should normally represent one main entity or concept.

4.4 Column

A column represents one property or attribute of an entity. In the Students table:

- StudentID represents the student's identification number.
- StudentName represents the student's name.
- Branch represents the student's branch.
- City represents the student's city.

Every column has a column name, a data type, and rules or constraints (when required). A column defines what kind of information can be stored in that part of the table. Columns are also sometimes called fields or attributes.

4.5 Row

A row represents one complete record in a table. Consider this row:

StudentID	StudentName	Branch	City
-----	-----	-----	-----

101 Aarav CSE Bengaluru

This row contains all the stored information about one student. In the complete Students table, the first row represents Aarav, the second row represents Bhavna, and the third row represents Charan. Rows are also called records.

4.6 Value

A value is the actual data stored at the intersection of a row and a column. In the record below:

StudentID StudentName Branch City

101 Aarav CSE Bengaluru

- 101 is the value under StudentID.
- Aarav is the value under StudentName.
- CSE is the value under Branch.
- Bengaluru is the value under City.

Each value belongs to one row and one column.

4.7 Complete Structure

Suppose a MySQL server contains a college application:

Level Example Purpose

MySQL Server Local MySQL Server Manages databases

Database / Schema CollegeManagement Contains related database objects

Table Students Stores information about students

Column StudentName Defines one property

Row Aarav's complete record Represents one student

Value Aarav Stores the actual data

Database and Table Are Not the Same

A database is the complete container, while a table stores data about one particular entity. For example, database CollegeManagement may contain the Students, Courses, Trainers, Attendance, and Marks tables. The database contains the tables --- a table does not contain the database.

Row and Column Are Not the Same

Row Column

Represents one complete record Represents one property

Moves horizontally across the table Moves vertically through the table

Example: one student Example: names of all students

Also called a record Also called a field or attribute

4.8 Mental Model

Think of a database as a collection of organized registers, kept in a cupboard:

Real-world object Database concept

The complete cupboard The database

The arrangement of registers The schema

Each register A table

Each heading in a register A column

Each filled entry A row

Each individual piece of information A value

■ Final Takeaway A database contains related data and database objects. A schema describes how the database is organized (in MySQL, database and schema usually refer to the same thing). A table stores data about one type of entity. A column represents one property. A row represents one complete record. A value is the actual data stored inside a table.

Interview Questions — Chapter 4

Q1. Database vs table?

Ideal answer: A database is a container for related database objects; a table stores records for one entity or concept.

Q2. Row vs column?

Ideal answer: A row is one complete record; a column is one property or attribute shared by the records.

Q3. What is a schema?

Ideal answer: A schema is the logical blueprint describing tables, columns, data types, relationships, and rules. In MySQL, database and schema are generally used interchangeably.

Q4. What is a value?

Ideal answer: A value is the actual piece of data stored at the intersection of a row and a column.

Q5. Can one database contain multiple tables?

Ideal answer: Yes. A college database can contain Students, Courses, Attendance, Marks, and Placements tables.